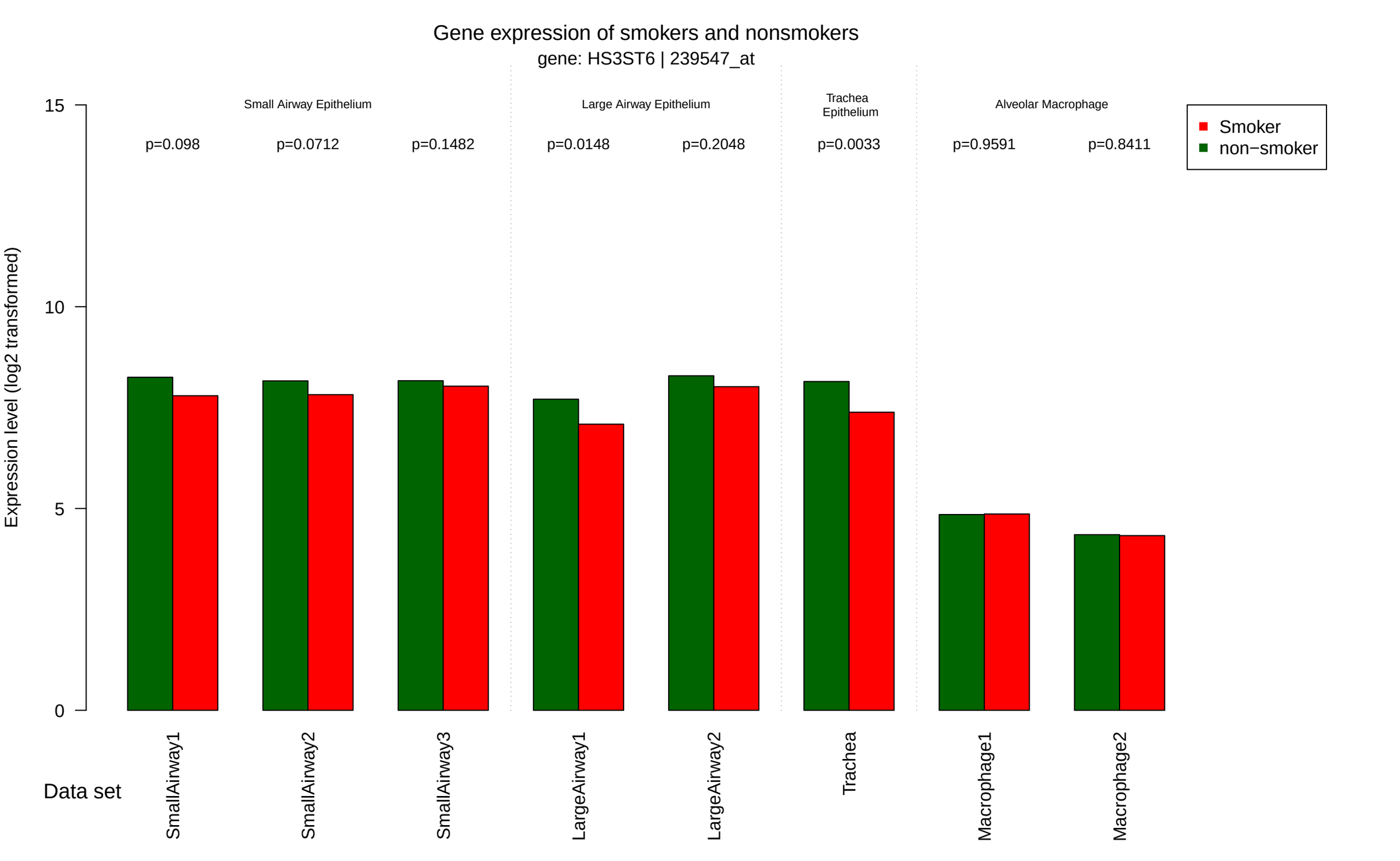
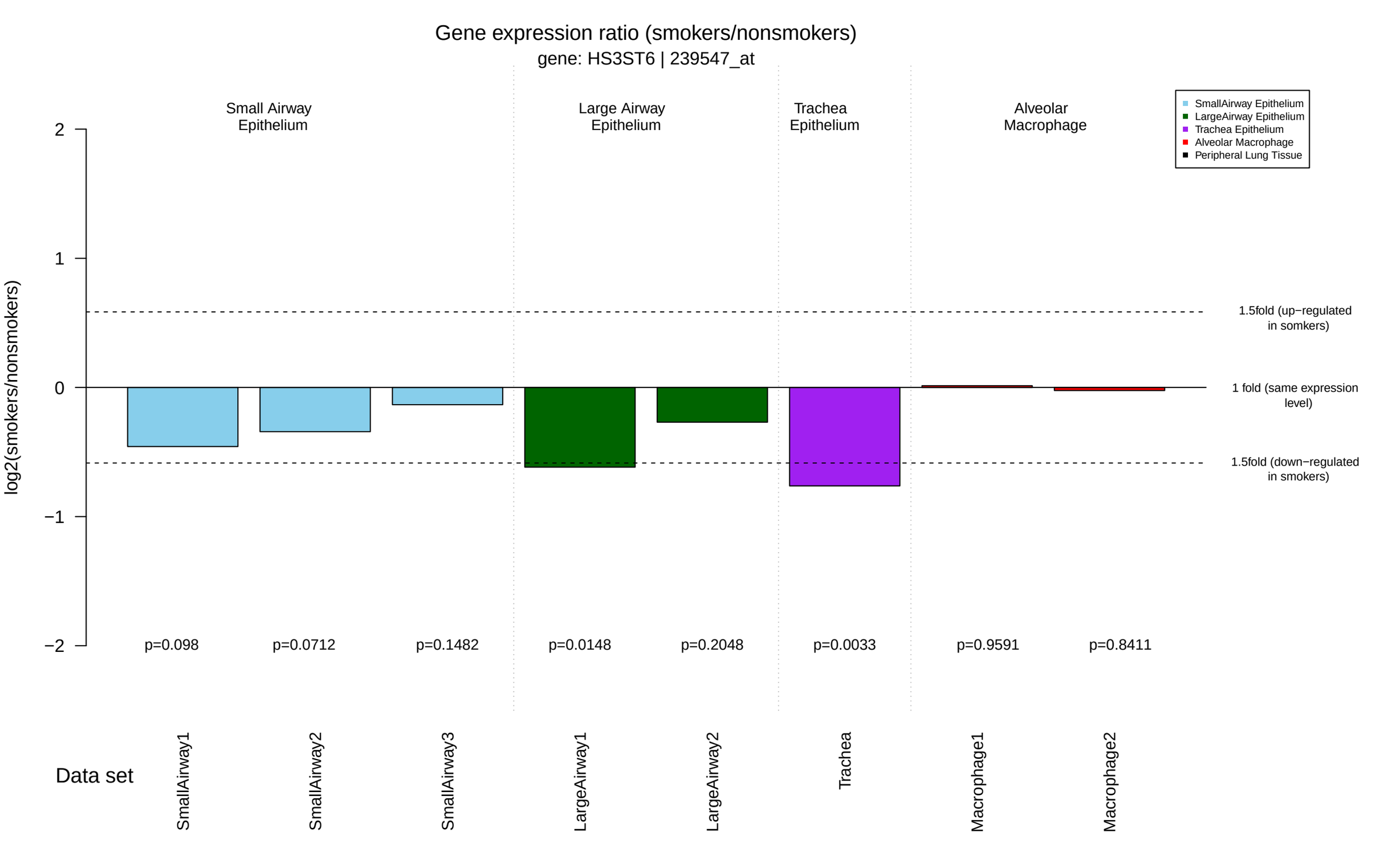
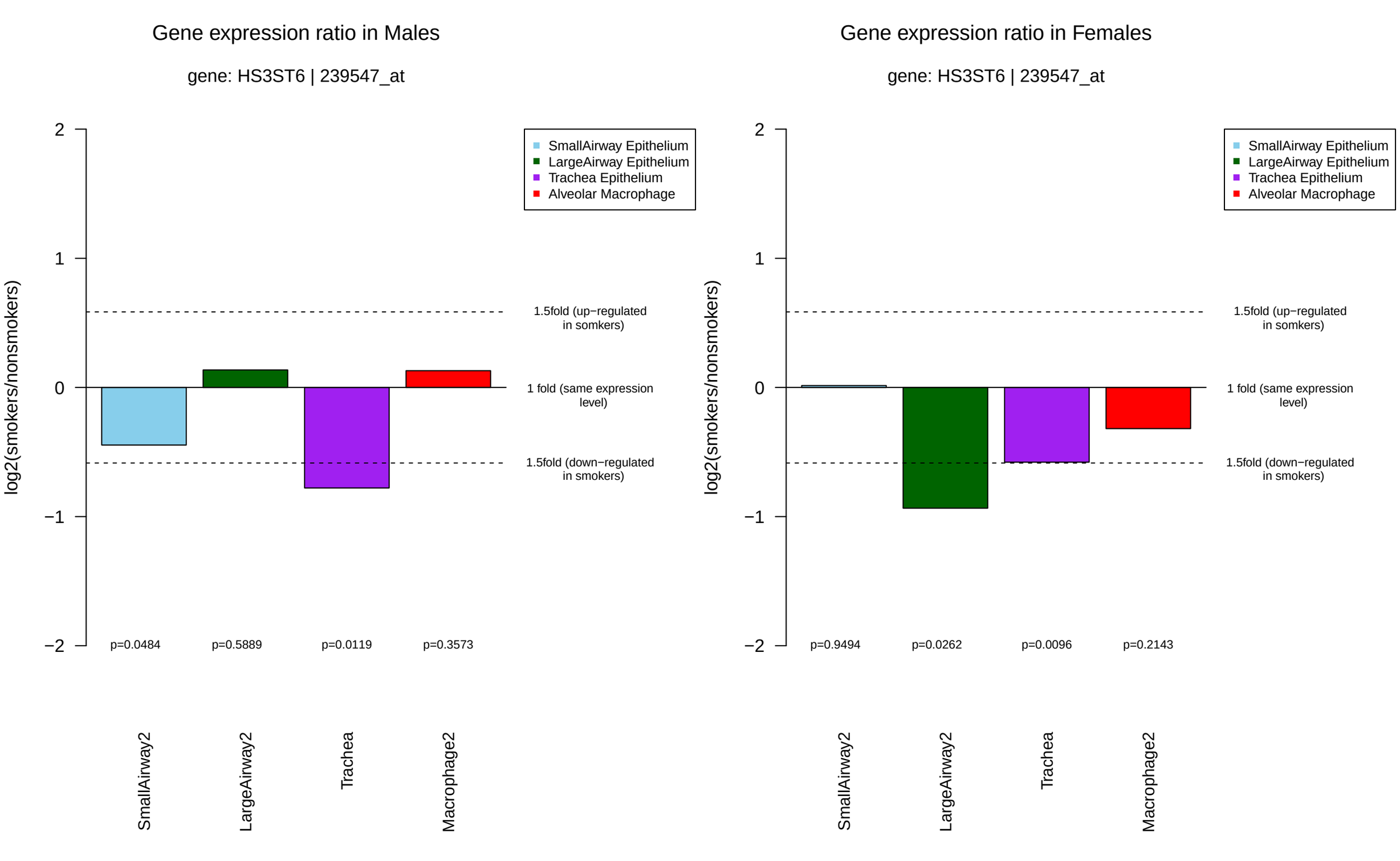
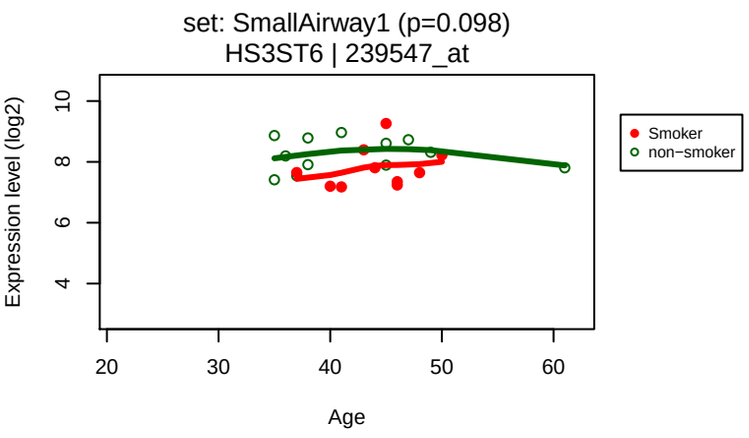
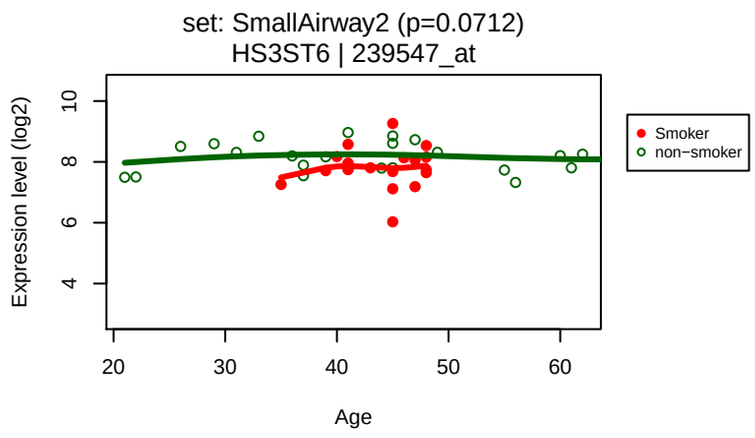
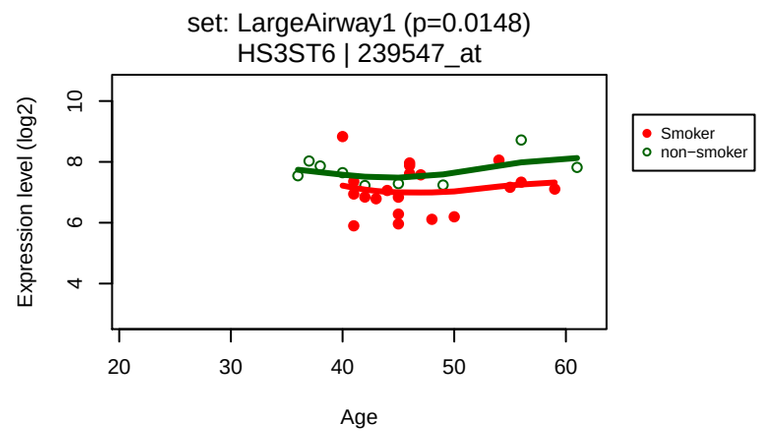

Gene expression of smokers and nonsmokers
gene: HS3ST6 | 239547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.098
p=0.0712
p=0.1482
p=0.0148
p=0.2048
p=0.0033
p=0.9591
p=0.8411
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

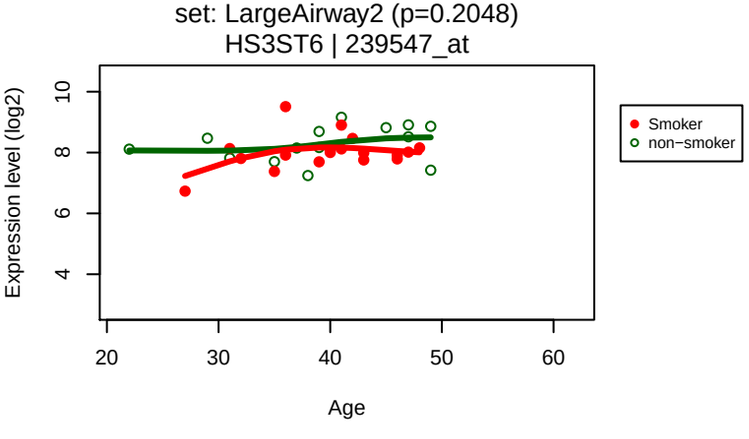

Gene expression ratio (smokers/nonsmokers)
gene: HS3ST6 | 239547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.098
p=0.0712
p=0.1482
p=0.0148
p=0.2048
p=0.0033
p=0.9591
p=0.8411
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

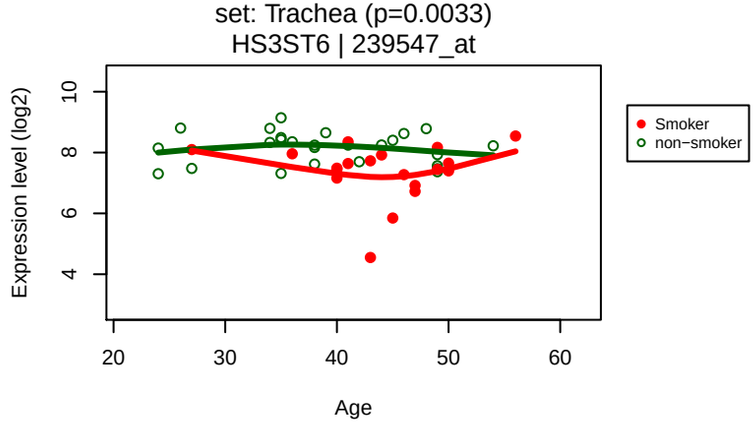

Gene expression ratio in Males
gene: HS3ST6 | 239547_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0484
p=0.5889
p=0.0119
p=0.3573
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HS3ST6 | 239547_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9494
p=0.0262
p=0.0096
p=0.2143
SmallAirway2
LargeAirway2
Trachea
Macrophage2

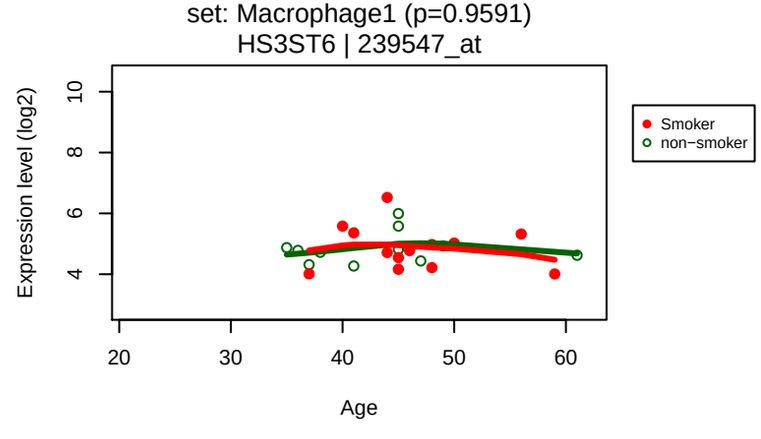

set: SmallAirway1 (p=0.098)
HS3ST6 | 239547_at
Expression level (log2)
Age
Smoker
non−smoker

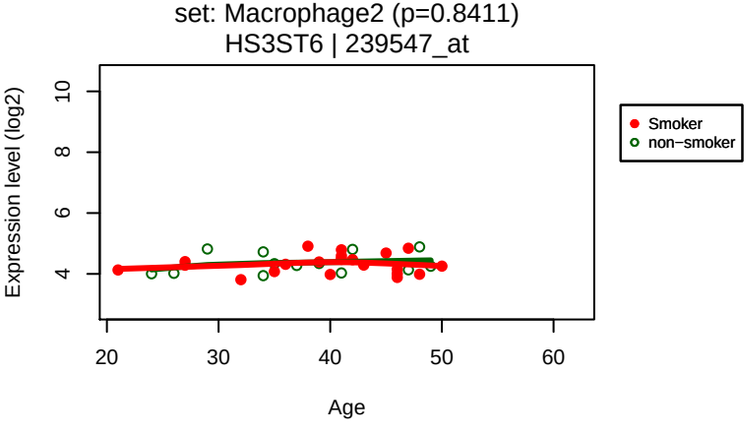

set: SmallAirway2 (p=0.0712)
HS3ST6 | 239547_at
Expression level (log2)
Age
Smoker
non−smoker

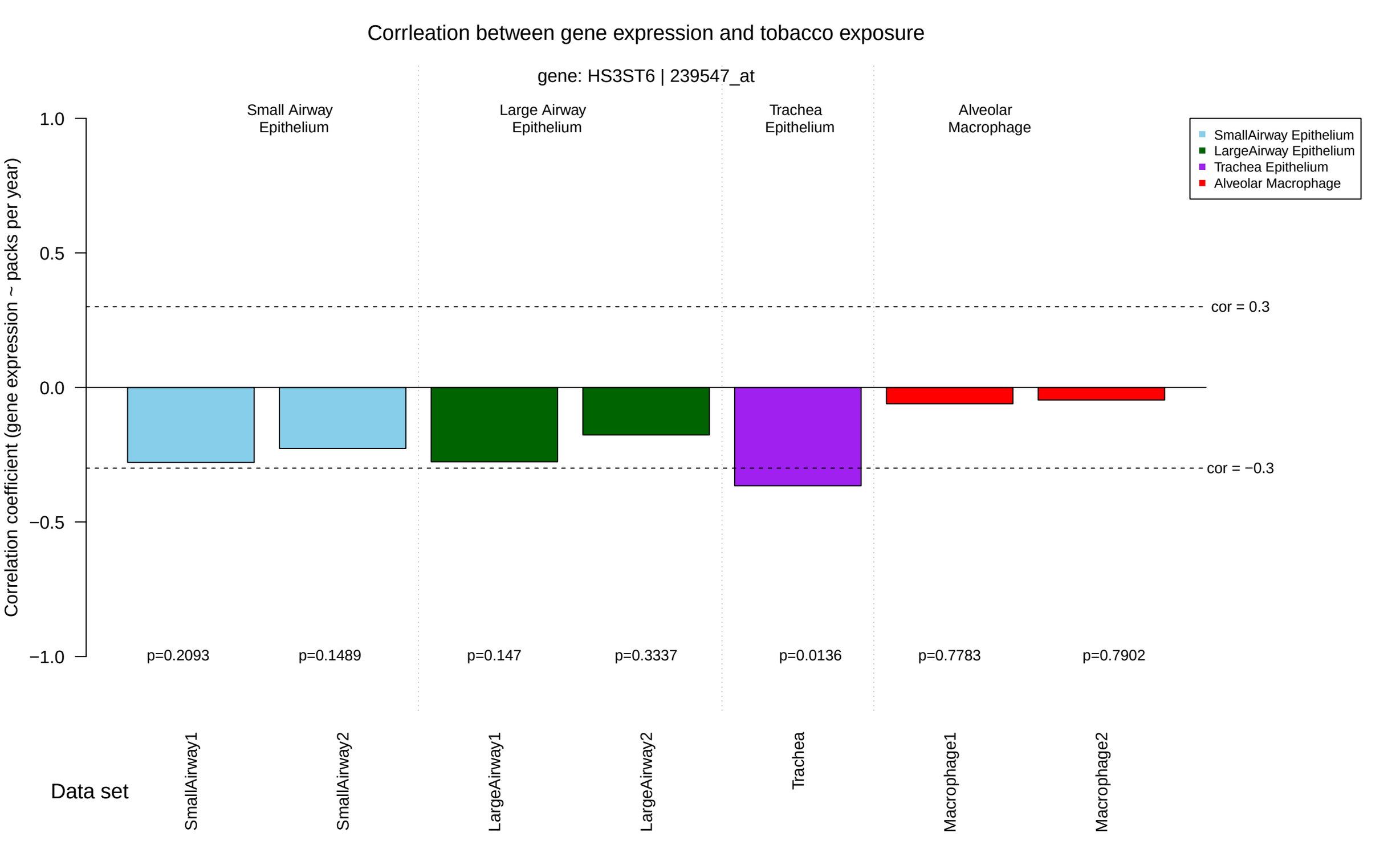

set: LargeAirway1 (p=0.0148)
HS3ST6 | 239547_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2048)
HS3ST6 | 239547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0033)
HS3ST6 | 239547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9591)
HS3ST6 | 239547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8411)
HS3ST6 | 239547_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HS3ST6 | 239547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2093
p=0.1489
p=0.147
p=0.3337
p=0.0136
p=0.7783
p=0.7902
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2